6. The Causal Calculus (do-calculus)

A consolidated, mathematical lecture companion

English notes based on the lecture slides *The Causal Calculus* (Prof. Sanghack Lee, GSDS SNU)

The central question is whether there is a systematic way to decide if $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ can be expressed in terms of the observational distribution $P(\mathbf{v})$. The truncated factorization sometimes yields identifiable effects (back-door, front-door) and sometimes not (the bow graph). Do-calculus (Pearl 1995) supplies three graphical inference rules, adding/removing observations, exchanging actions and observations, adding/removing actions, which are *complete*: if an effect is identifiable, do-calculus derives it.

These notes consolidate repeated slides, write every derivation line by line with the graph used to justify each rule, redraw the catalog of identifiable and non-identifiable graphs, and give an answer to the closing exercise. Physical PDF page numbers are listed in Section 7.

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1 Three examples via the truncated factorization

Purpose. See how identification works, and fails, when the **U**-terms of a semi-Markovian model must be eliminated by hand.

Theorem 1.1 (Generalized truncated factorization)**.** *In a semi-Markovian model $\mathcal{M}$, the distribution generated by* $\mathrm{do}(\mathbf{X}=\mathbf{x})$ *is*

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V}\setminus\mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}),$$
$$\mathbf{V}' = \mathbf{V} \setminus \mathbf{X},$$

and for $\mathbf{Y} \subseteq \mathbf{V}\setminus\mathbf{X}$, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v}\setminus(\mathbf{x}\cup\mathbf{y})} \sum_{\mathbf{u}} \prod_{V_i\in\mathbf{V}\setminus\mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$.

Identification succeeds when the right-hand side can be rewritten without **U**.

1.1 Back-door graph

$Z \to X$, $Z \to Y$, $X \to Y$ with independent U_Z, U_X, U_Y. In $\mathcal{M}_x$: $Z \leftarrow f_Z(u_Z)$, $X \leftarrow x$, $Y \leftarrow f_Y(x, z, u_Y)$.

$$\begin{aligned} P(\mathbf{v}' \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) \\ &= \Big(\sum_{u_Z} P(z \mid u_Z) P(u_Z)\Big)\Big(\sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y)\Big)\Big(\sum_{u_X} P(u_X)\Big) \\ &= P(z) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y \mid x, z) = P(z) P(y \mid x, z), \end{aligned}$$

using $U_Y \perp\!\!\!\perp \{X, Z\}$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z) P(z)$.

1.2 Bow graph

$X \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Y \leftarrow f_Y(x, u, u_Y)$.

$$P(y \mid \mathrm{do}(x)) = \sum_u P(u) \sum_{u_Y} P(y \mid x, u, u_Y) P(u_Y) \sum_{u_X} P(u_X) = \sum_u P(y \mid x, u) P(u).$$

U cannot be eliminated: $P(u)$ and $P(y \mid x, u)$ are not functions of $P(x, y)$. (Observationally, $P(\mathbf{v}) = \sum_u P(x \mid u, u_X) P(y \mid x, u, u_Y) P(\mathbf{u})$ weights u by $P(u \mid x)$ instead.)

1.3 Front-door graph

$X \to Z \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Z \leftarrow f_Z(x, u_Z)$, $Y \leftarrow f_Y(z, u, u_Y)$.

$$\begin{aligned}
P(z, y \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid x, u_Z) P(y \mid z, u, u_Y) P(\mathbf{u}) \\
&= \Big(\sum_{u_Z} P(z \mid x, u_Z) P(u_Z) \Big) \Big(\sum_{u, u_Y} P(y \mid z, u, u_Y) P(u, u_Y) \Big) \\
&= P(z \mid x) \sum_{u} P(y \mid z, u) \sum_{x'} P(u \mid x') P(x') \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x') P(x') && (Y \perp\!\!\!\perp X \mid Z, U) \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x', z) P(x') && (U \perp\!\!\!\perp Z \mid X) \\
&= P(z \mid x) \sum_{x'} \sum_{u} P(y, u \mid z, x') P(x') = P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').
\end{aligned}$$

Therefore the **front-door formula**

$$P(y \mid \mathrm{do}(x)) = \sum_{z} P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').$$

2 Identification approaches

Purpose. State the syntactic goal that do-calculus serves.

In both the back-door and front-door settings the goal was to reduce $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to an expression whose terms are estimable from $P(\mathbf{v})$. One way is to write the truncated factorization and derive a $\mathbf{U}$-free expression, as above. Another is to rewrite the query into a *do-free* expression without ever mentioning $\mathbf{U}$. We seek rules (axioms) that transform do-expressions into do-free ones while preserving equality with the target effect.

3 Do-calculus

Purpose. Derive the three rules from three insights and explain their graphical side conditions.

3.1 Prelude: independence in a submodel

$\mathrm{do}(\mathbf{X} = \mathbf{x})$ replaces f_X by the constant x for each $X \in \mathbf{X}$. Graphically $\mathcal{G}_{\overline{\mathbf{X}}}$ represents $\mathbf{X}$ not influenced by any other variable. Moreover $\mathbf{X}$ are now constants, which corresponds to conditioning on $\mathbf{X} = \mathbf{x}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. Hence

$$(\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}) \text{ under } \mathrm{do}(\mathbf{x}) \quad \Longleftarrow \quad (\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C}, \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}},$$

equivalently $(\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C})$ in $\mathcal{G} \setminus \mathbf{X}$ (the graph with $\mathbf{X}$ deleted) when $\mathbf{X}$ is disjoint from $\mathbf{A}, \mathbf{B}, \mathbf{C}$. For example, with $A \to X \to D$, the intervention severs A from X, and $\mathbf{X}$ fixed blocks every path through X.

3.2 Insight 1: adding/removing observations

Graph: $Z \to X \to Y$, $X \leftrightarrow Y$ (via U). Then

$$P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, u_Y, u) P(\mathbf{u}) = P(z) \sum_{u} P(y \mid x, u) P(u).$$

Marginalizing, $P(z \mid \mathrm{do}(x)) = P(z)\sum_u P(u)\sum_y P(y \mid x,u) = P(z)$, so

$$P(y \mid \mathrm{do}(x), z) = \frac{P(y,z \mid \mathrm{do}(x))}{P(z \mid \mathrm{do}(x))} = \sum_u P(y \mid x,u)P(u) = \sum_z P(y,z \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x)).$$

The observation z can be dropped, and the graphical reason is $(Z \perp_d Y \mid X)_{\mathcal{G}_{\overline{X}}}$. In the original model Z and Y are inseparable: $Z \not\perp\!\!\!\perp Y$ (through $Z \to X \to Y$) and $Z \not\perp\!\!\!\perp Y \mid X$ (X is a collider on $Z \to X \leftrightarrow Y$). In the $\mathrm{do}(X)$-world $\mathcal{M}_x$, however, the edge $Z \to X$ is cut and Y, Z are d-separated; since $P(y \mid \mathrm{do}(x), z) = P_x(y \mid z)$ is an ordinary conditional in the submodel, the observation can be removed.

3.3 Insight 2: exchanging action and observation

Graph: $Z \to X$, $Z \to Y$, $X \to Y$. After observing Z, Y reacts to X in the same way with and without intervention: given Z, Y is associated with X only through causal paths, so *seeing* $X = x$ is equivalent to *doing* $X = x$. If Z blocks all back-door paths, the remaining association is causation:

$$(Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}} \implies P(y \mid \mathrm{do}(x), z) = P(y \mid x, z).$$

Check: $P(y,z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z)P(y \mid x, z, u_Y)P(\mathbf{u}) = P(z)P(y \mid x, z)$ and $P(z \mid \mathrm{do}(x)) = \sum_y P(z)P(y \mid x, z) = P(z)$, so $P(y \mid \mathrm{do}(x), z) = P(z)P(y \mid x, z)/P(z) = P(y \mid x, z)$.

3.4 Insight 3: adding/removing actions

Graph: $X \to Y$, $Z \leftrightarrow Y$ (latent U_{zy}), $Z \leftrightarrow X$ (latent U_{zx}). If there is no causal path from X to Z, intervening on X has no effect on Z:

$$(Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \implies P(z \mid \mathrm{do}(x)) = P(z).$$

Check: $P(z \mid \mathrm{do}(x)) = \sum_y \sum_{u_{zy},u_{zx}} P(z \mid u_{zy}, u_{zx})P(y \mid x, u_{zy})P(u_{zy}, u_{zx}) = \sum_{u_{zy},u_{zx}} P(z \mid u_{zy}, u_{zx})P(u_{zy}, u_{zx}) = P(z)$.

3.5 The rules

Notation: $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ is $\mathcal{G}$ with edges into $\mathbf{X}$ and out of $\mathbf{Z}$ removed.

Theorem 3.1 (Rules of do-calculus; Pearl 1995)**.** *For any do-distribution induced by a causal model $\mathcal{M}$ with diagram $\mathcal{G}$, and disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$:*

Rule 1 *(adding/removing observations)* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ *if* $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Rule 2 *(action/observation exchange)* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w})$ *if* $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$.

Rule 3 *(adding/removing actions)* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ *if* $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}}$, *where* $\mathbf{Z}(\mathbf{W}) = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Unconditional versions ($\mathbf{W} = \emptyset$) read: Rule 1 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, Rule 2 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$, Rule 3 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}$.

Simplified form. Let $\mathcal{H} = \mathcal{G} \setminus \mathbf{X}$ and $Q = P_{\mathbf{x}}$. Then

$$\begin{aligned}
&\text{R1: } Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}},\\
&\text{R2: } Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\underline{\mathbf{Z}}}},\\
&\text{R3: } Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\overline{\mathbf{Z(W)}}}},\ \mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{H}}.
\end{aligned}$$

Every rule is a statement about one fixed submodel $\mathcal{M}_{\mathbf{x}}$; $\mathrm{do}(\mathbf{x})$ is just carried along.

3.6 Understanding Rule 2

One may add the observation $\mathbf{x}$ under $\mathrm{do}(\mathbf{x})$: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{x}, \mathbf{z}, \mathbf{w})$. So Rule 2 asks whether the model and the submodel generate the same conditional probability, e.g. (with no outer $\mathrm{do}(\mathbf{x})$) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{z}, \mathbf{w})$.

Regime indicator. Augment $\mathcal{G}$ (here $W \to Z$, $W \to Y$, $Z \to Y$) with a parent $F_Z \to Z$ selecting the mechanism:

$$f'_Z = \begin{cases} f_Z(W, U_Z) & \text{if } F_Z = 0,\\ z & \text{if } F_Z = 1. \end{cases}$$

Then the Rule 2 equality $P(y \mid \mathrm{do}(z), z, w) = P(y \mid z, w)$ reads $P(y \mid z, w, F_Z = 1) = P(y \mid z, w, F_Z = 0)$, i.e. $(F_Z \perp\!\!\!\perp Y \mid Z, W)$ in the augmented graph $\mathcal{G}'$. Since Z is given, every d-connecting path from F_Z must be $F_Z \to Z \leftarrow \cdots Y$ (the collider Z is opened); so we check paths that leave Z through an arrowhead, i.e. back-door paths, which is the same as checking Z versus Y in $\mathcal{G}_{\underline{Z}}$.

3.7 Understanding Rule 3

"Changing a function" subsumes "changing it to a constant", so

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), F_Z = 1, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), F_Z = 0, \mathbf{w}),$$

which holds if $F_{\mathbf{Z}} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ in $\mathcal{G}'$. A path leaving F_Z has the form $F_Z \to Z \to \cdots$ (always feasible) or $F_Z \to Z \leftarrow \cdots$, feasible only if the collider Z is opened, i.e. $Z \in An(\mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. Hence edges into Z can be cut exactly for $Z \in \mathbf{Z(W)}$, giving the condition $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}}$.

Example 3.2 (Why $\mathbf{Z(W)}$ is needed)**.** Graph: $T \to Z \to Y$, $Z \to W$. Is $P(t \mid \mathrm{do}(z), w) = P(t \mid w)$? In $\mathcal{G}_{\overline{Z}}$, T is isolated, so a naive check $(T \perp_d Z \mid W)_{\mathcal{G}_{\overline{Z}}}$ would say yes. But $Z \in An(W)$, so $\mathbf{Z(W)} = \emptyset$ and the correct graph is $\mathcal{G}$ itself, where $T \to Z$ remains and the condition fails. Indeed, in almost every compatible model,

$$P(t \mid \mathrm{do}(z), w) = \frac{P(t, w \mid \mathrm{do}(z))}{P(w \mid \mathrm{do}(z))} = \frac{\sum_y P(t)P(w \mid z)P(y \mid z)}{\sum_{y,t} P(t)P(w \mid z)P(y \mid z)} = \frac{P(t)P(w \mid z)}{P(w \mid z)} = P(t) \neq P(t \mid w),$$

since W carries information about T through Z observationally.

4 Do-calculus and identification

Purpose. State completeness, work the front-door derivation, and give tools for proving non-identifiability.

Theorem 4.1 (Completeness; Huang and Valtorta 2006, Shpitser and Pearl 2006)**.** *$Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $P(\mathbf{v})$ and $\mathcal{G}$ iff there is a finite sequence of applications of the rules of do-calculus and the probability axioms that reduces Q to a do-free expression.*

Syntactic goal: re-express Q without do.

4.1 Front-door by do-calculus

Smoking $S \to$ Tar $T \to$ Cancer C, with $S \leftrightarrow C$.

$$\begin{aligned}
P(c \mid \mathrm{do}(s)) &= \sum_t P(c \mid \mathrm{do}(s), t) P(t \mid \mathrm{do}(s)) && \text{probability axioms} \\
&= \sum_t P(c \mid \mathrm{do}(s), \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\overline{S}\underline{T}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 3: } (S \perp_d C \mid T)_{\mathcal{G}_{\overline{T}\,\overline{S}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid s) && \text{Rule 2: } (T \perp_d S)_{\mathcal{G}_{\underline{S}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', \mathrm{do}(t)) P(s' \mid \mathrm{do}(t)) && \text{probability axioms} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s' \mid \mathrm{do}(t)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\underline{T}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s') && \text{Rule 3: } (T \perp_d S)_{\mathcal{G}_{\overline{T}}}.
\end{aligned}$$

Justifications: in $\mathcal{G}_{\overline{S}\underline{T}}$ the only edges are $S \leftrightarrow C$ and $S \to T$, and S is given; in $\mathcal{G}_{\overline{T}\,\overline{S}}$, S is disconnected except through $S \leftrightarrow C$, which T does not open; in $\mathcal{G}_{\underline{S}}$, $S \to T$ is removed and $S \leftrightarrow C \leftarrow T$ is blocked at the collider C; in $\mathcal{G}_{\underline{T}}$, $T \leftarrow S \leftrightarrow C$ is blocked by S; in $\mathcal{G}_{\overline{T}}$, T has no edges to S except via the collider C.

4.2 Proving non-identifiability

Example 4.2. Bow graph $X \to Y$, $X \leftrightarrow Y$, U a fair coin, $\mathcal{F}$: $X \leftarrow U$, $Y \leftarrow f_Y(X, U)$. Define two models that agree with $\mathcal{F}$ whenever $U = X$ (the only case ever observed) but differ off the diagonal:

$$\mathcal{F}_x^{(1)} : \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 0 & \text{otherwise} \end{cases},$$

$$\mathcal{F}_x^{(2)} : \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 1 & \text{otherwise.} \end{cases}$$

Under $\mathrm{do}(X = x)$, U and X need not match:

$$P^{(i)}(Y = 1 \mid \mathrm{do}(x)) = \sum_u P^{(i)}(y \mid x, u) P(u) = P(y \mid x) P(x) + \mathbb{1}_{i=2}\,(1 - P(x)).$$

Both models induce $\mathcal{G}$ and the same $P(\mathbf{v})$, but the effects differ whenever $P(x) < 1$.

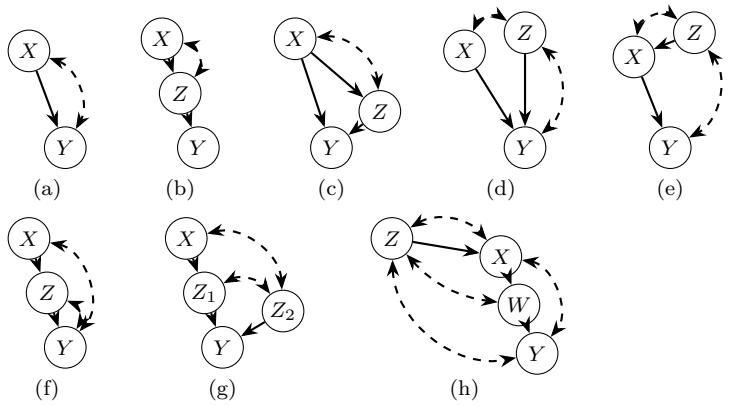
Lemma 4.3 (Graph–subgraph; Tian and Pearl 2002)**.** *If $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is not identifiable in $\mathcal{G}$, it is not identifiable in any graph obtained by adding a directed or bidirected edge to $\mathcal{G}$. Conversely, if Q is identifiable in $\mathcal{G}$, it remains identifiable after removing edges.*

(Adding an edge only enlarges the class of compatible models, since a graph permits but does not force dependence.)

Theorem 4.4 (Non-identifiability criterion, singleton; Tian and Pearl 2002)**.** *If there is a bidirected path connecting X to any of its children in $\mathcal{G}$, then $P(\mathbf{v}' \mid \mathrm{do}(x))$ is not identifiable from $P(\mathbf{v})$ and $\mathcal{G}$.*

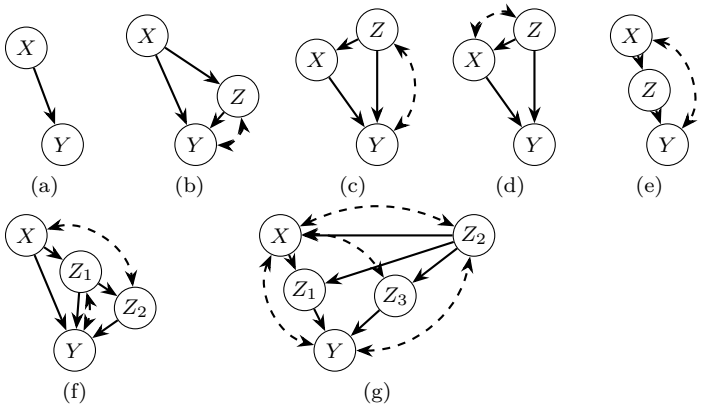
4.3 Catalogue of graphs

Graphs in which $P(y \mid \mathrm{do}(x))$ is not identifiable (Pearl 2000, Fig. 3.9):



(a) is the bow graph; (b), (c) contain the bow $X \to Z$, $X \leftrightarrow Z$ on the way to Y; in (d)–(h) X is connected to its child by a bidirected path (e.g. in (d) $X \leftrightarrow Z \leftrightarrow Y$, in (g) $X \leftrightarrow Z_2 \leftrightarrow Z_1$ with Z_1 a child of X).

Graphs in which $P(y \mid \mathrm{do}(x))$ is identifiable (Pearl 2000, Fig. 3.8):



For instance in (b) X has no incoming edges, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$;

(c) and (d) are identified by back-door adjustment for Z (in (d) the confounding between X and Z is harmless), and (e) by the front-door formula.

5 Back to Simpson's paradox

Purpose. Use the do-operator to dissolve the paradox and explain why statistics alone cannot resolve it.

5.1 Pearson's shock

Pearson, Lee and Bramley-Moore (1899) observed that mixing heterogeneous groups, each of which shows no "organic" correlation, can produce a correlation, which they called *spurious*; since no community is perfectly homogeneous, correlations are always liable to an error of unknown size, a "shock" to those who equate correlation with causation. Within each subgroup A and

B may be uncorrelated, yet in the mixture a correlation appears: an artefact of heterogeneity. The same data can lead to opposite conclusions depending on whether we aggregate or stratify. This is the essence of Simpson's paradox, and it cannot be resolved by statistics alone; it requires causal reasoning.

Examples. (1) UC Berkeley's alleged sex bias in graduate admissions (Bickel et al., *Science* 1975): overall a higher admission rate among male applicants (44% vs. 30%), but within departments a slight bias in favor of women. Women tended to apply to more competitive departments, a societal pattern rather than discrimination by committees. But is conditioning on department the right thing to do? That depends on the causal model, e.g. on latent confounding between department choice and admission.

(2) "Reverse regression" (1970s–80s): in salary-discrimination cases, should one compare salaries of equally qualified men and women, or qualifications of equally paid men and women? (Pearl 2009, §4.5; Bhadane et al. 2025.)

5.2 Dissolving the paradox with the do-operator

The tables of Lecture 1 give

$$P(Y \mid F, X) < P(Y \mid F, \neg X), \qquad P(Y \mid \neg F, X) < P(Y \mid \neg F, \neg X), \qquad P(Y \mid X) > P(Y \mid \neg X).$$

If instead we have, in do-notation,

$$P(Y \mid F, \mathrm{do}(X)) < P(Y \mid F, \mathrm{do}(\neg X)), \qquad P(Y \mid \neg F, \mathrm{do}(X)) < P(Y \mid \neg F, \mathrm{do}(\neg X)),$$

and the action does not affect F (sex), then necessarily $P(Y \mid \mathrm{do}(X)) < P(Y \mid \mathrm{do}(\neg X))$. Proof:

$$P(y \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f),$$

by Rule 3 (F is not a descendant of X). Both sides are averages of the stratum-specific effects with the *same* weights $P(f)$, so the inequality is preserved.

5.3 The paradox stems from causal interpretation

Surprise surfaces only when we speak about efficacy, not about evidence for prediction. When two causal models generate the same data and one recommends the drug while the other does not, the decision must be driven by causal considerations; no statistical criterion warns us against consulting the wrong table.

Theorem 5.1 (Sure-thing principle; Savage 1972, Pearl 2000 Thm. 6.1.1)**.** *An action X that increases the probability of an event Y in each subpopulation must also increase the probability of Y in the population as a whole, provided that the action does not change the distribution of the subpopulations.*

Any statistical relationship between two variables can be reversed by including additional factors in the analysis; people think in terms of causes, not proportions. Reversal is possible in the calculus of proportions (prediction), impossible in the calculus of causation. Applications include the adjustment problem, sex and salary discrimination cases, the design of optimal treatments and fairness in machine learning.

5.4 Appendix: the Berkeley data

Department	Men applied	Men admitted	Women applied	Women admitted
A	825	62%	108	82%
B	560	56%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
Total	2691	44%	1835	30%

Women applied disproportionately to competitive departments (C–F). Which causal graph is appropriate?

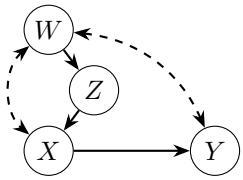
- $\mathcal{G}_1$: Sex $\to$ Dept $\to$ Admit, Sex $\to$ Admit (Dept is a mediator). The total effect $P(\text{admit} \mid \text{do(sex)})$ is the aggregate rate; adjusting for Dept yields only the (controlled) direct effect. The relevant questions become: *direct effect*, is there discrimination within departments? *indirect effect*, does sex influence which department one applies to?
- $\mathcal{G}_2$: $U \to$ Dept, $U \to$ Sex, Dept $\to$ Admit, Sex $\to$ Admit, with U latent (Dept is a confounder via U). Then Dept blocks the back-door path Sex $\leftarrow U \to$ Dept $\to$ Admit, and back-door adjustment gives $P(\text{admit} \mid \text{do(sex)}) = \sum_d P(\text{admit} \mid \text{sex}, d)P(d)$, which reveals no significant discrimination within departments.

Same data, same adjustment, but different causal interpretations.

6 Summary and food for thought

- Do-calculus gives a syntactic characterization of policy evaluation for atomic interventions.
- The problem of confounding and identification is essentially solved non-parametrically.
- Simpson's paradox is mathematised and dissolved.
- Applications pervade the social and health sciences, statistics, ML and AI.

Food for thought: the napkin graph. $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$. Is $P(y \mid \text{do}(x))$ identifiable?



Yes, although neither back-door nor front-door applies. Intervening on Z makes X an unconfounded treatment in the subpopulation defined by W:

$$P(y \mid \text{do}(x)) = \frac{\sum_w P(x, y \mid z, w)P(w)}{\sum_w P(x \mid z, w)P(w)} \qquad \text{(for any } z\text{)}.$$

(Derivation: $P(y \mid \text{do}(x)) = P(y \mid \text{do}(x), \text{do}(z))$ by Rule 3; $P(x, y \mid \text{do}(z)) = \sum_w P(x, y \mid z, w)P(w)$ by back-door through W; and $P(y \mid \text{do}(z), \text{do}(x)) = P(y \mid \text{do}(z), x)$ by Rule 2, since in $\mathcal{G}_{\overline{Z}\underline{X}}$ the only paths from X to Y go through W, which is cut from X only via Z. Lecture 7 obtains this systematically.)

7 Source coverage

The source deck has 77 physical pages (44 numbered slides with build-ups).

Topic	Physical pages
Motivation, generalized truncated factorization, three examples	1–12
Identification approaches	13–17
Prelude, three insights	18–32
Rules of do-calculus and their interpretation	33–49
Completeness, front-door derivation, non-identifiability, graph catalog	50–58
Simpson's paradox, sure-thing principle, summary, napkin	59–73
Appendix: Berkeley admissions, references	74–77